# PHYSICS TECHNE SYMPOSIUM HACKATHON

TEAM ΨΦ

-Aaditya Arora(25JE0777)

-Karunesh Singh Adhikari(25JE1046)

-Alugubilli Sai Raghava(25JE0780)

-Nitin Singh(25JE0766)

# MISSION SCENARIO: VOYAGER-X SIGNAL RECOVERY

- The deep-space probe **Voyager-X**, currently executing a gravity-assist maneuver around Jupiter, has experienced a critical failure in its high-gain antenna system. As a result, the spacecraft has lost its primary communication link with Earth and has gone silent.
- However, the Deep Space Network (DSN) ground station at Chandrapur has detected a faint 5-minute burst of radio transmission from the probe's backup low-gain beacon. The captured signal is extremely weak and barely distinguishable from cosmic background noise.

# OUR MISSION

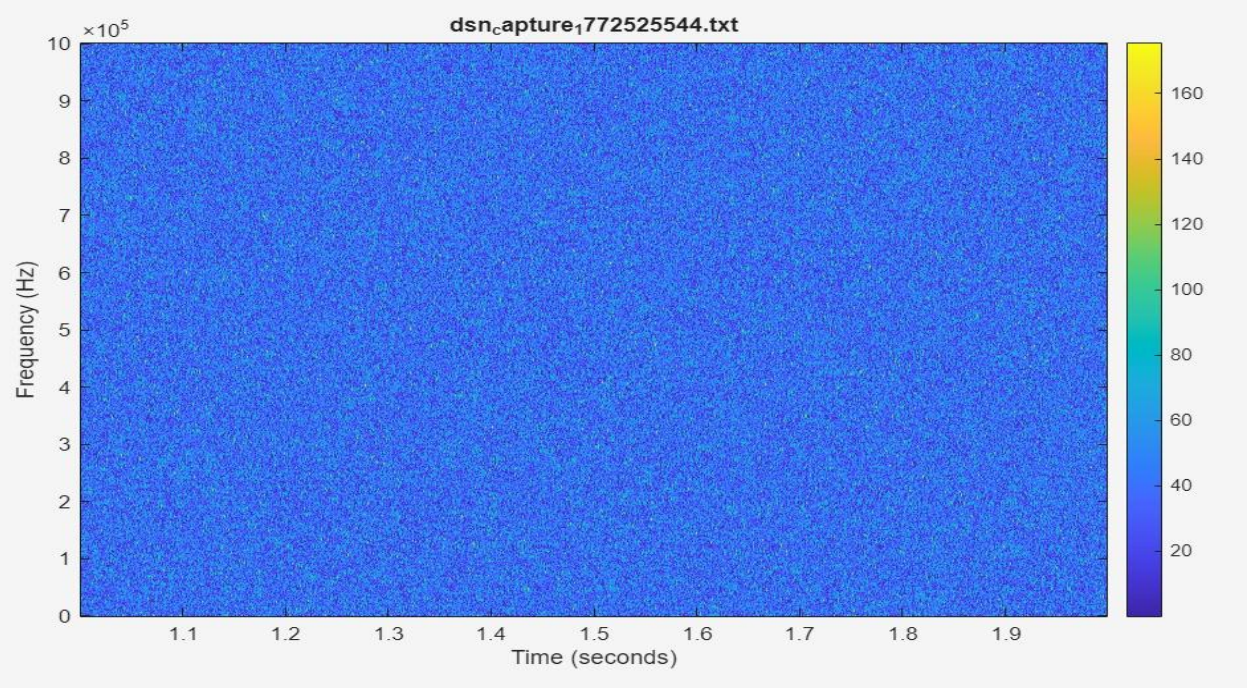
- We are acting as **Ground Station Engineers** responsible for recovering the probe's transmission. We have been provided with a fragmented raw **IQ (In-Phase and Quadrature)** recording of the radio spectrum. Our task is to design and implement a **software processing pipeline** to analyze the signal and recover all meaningful telemetry data transmitted by the probe.

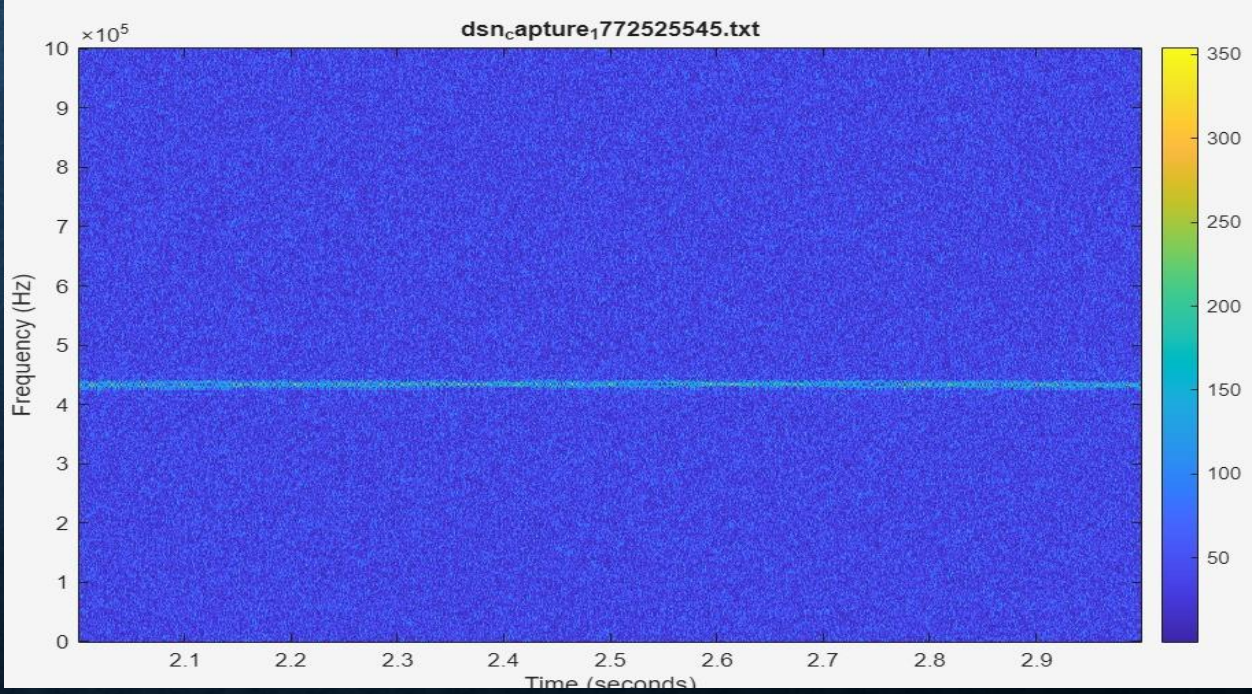
# OUR APPROACH TO SIGNAL RECOVERY

- From the given input dataset, we made C sharp script to convert the given hexadecimal values into single precision floating point values(eg: 0x3F2E6E24,0x3FAB851B -> 0.6813681,1.3399996). The received floating point pairwise value will be treated as coordinates corresponding to In-phase and Quadrature values respectively. Since we received a massive dataset of approx 15 gb(200files containing 2 million samples), we wrote a matlab script to scan the spectrogram for every second, we are passing a complete file and checking whether we are getting signal for each file. Upon scanning the spectrogram, we found out that signal started to emerge from 3$^{rd}$ file and end at the 71$^{st}$ file. So now for further procedure, we will use only these 3$^{rd}$ -71$^{st}$ files

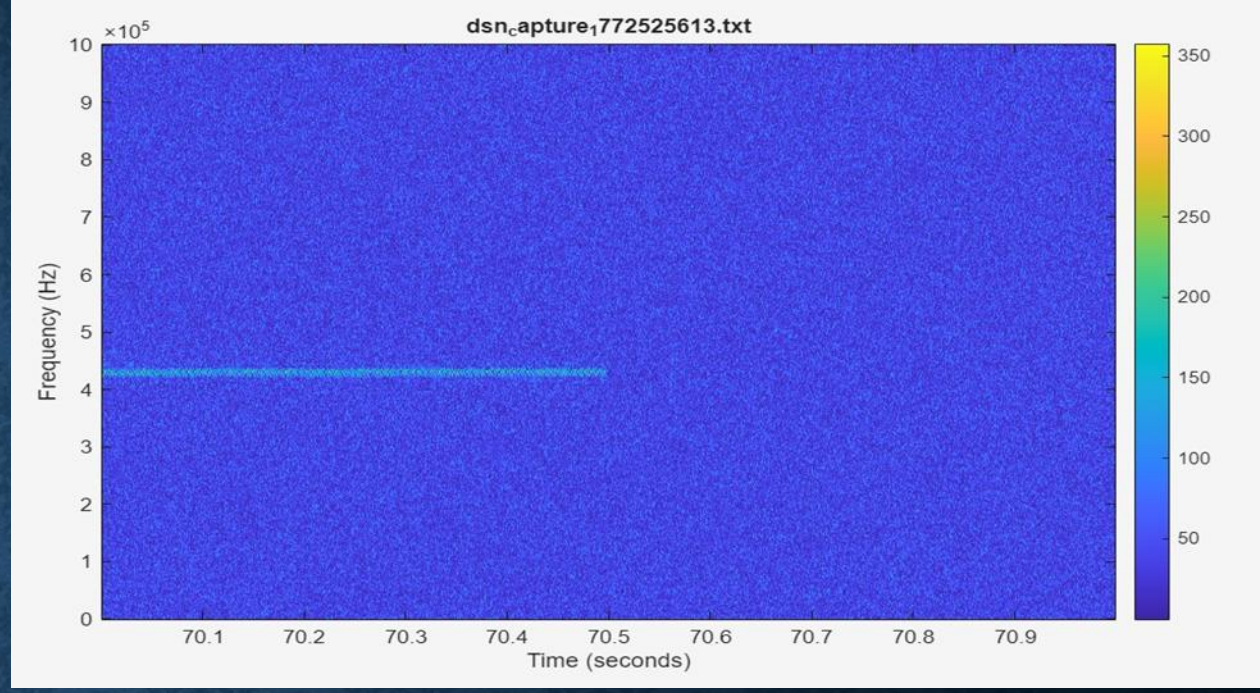
# Spectrogram Analysis

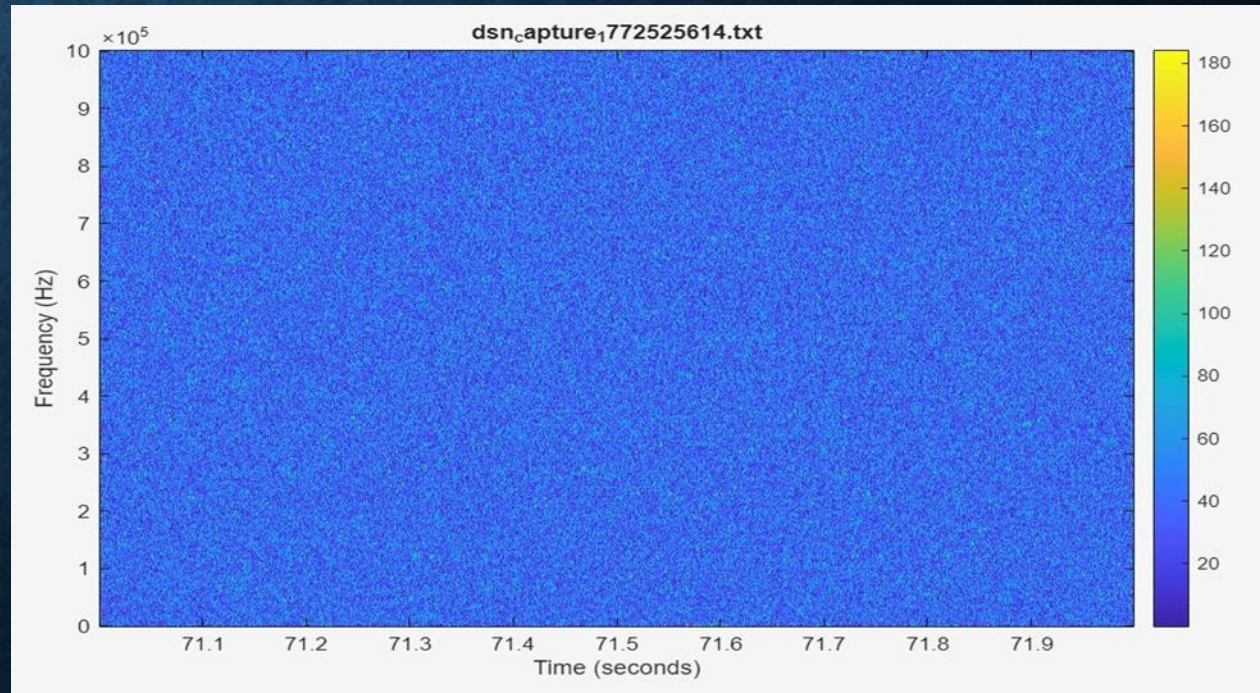
The spectrogram analysis shows that no significant signal was detected in Files 1 and 2. However, starting from File 3, a clear signal begins to appear. The detected signal is centered around approximately **432 kHz**, indicating the likely presence of the probe's transmission.



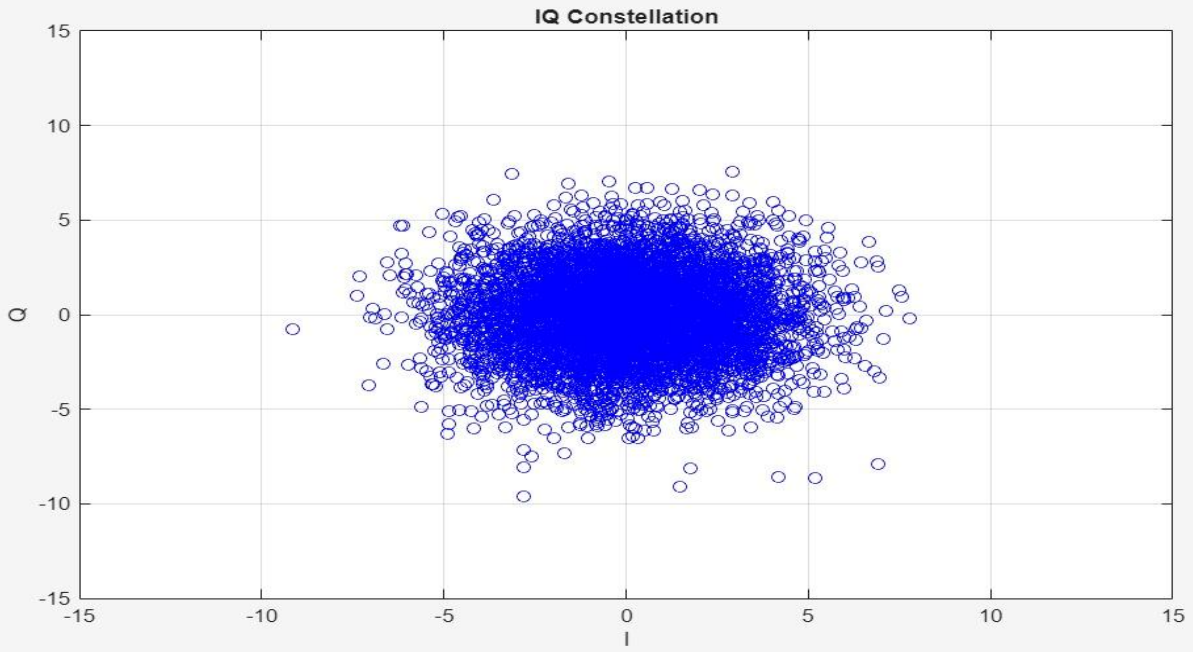



Analysis of the spectrogram shows that the signal persists until File 71 and ends at approximately 70.5 seconds from the start of the data. Since MATLAB loads the spectrogram for each file at one-second intervals, the 70.5-second mark corresponds to a point midway through the analysis of File 71. The signal is centered around ~430 kHz and exhibits a frequency drift of a few kilohertz over time. No detectable signal is observed in the remaining files.

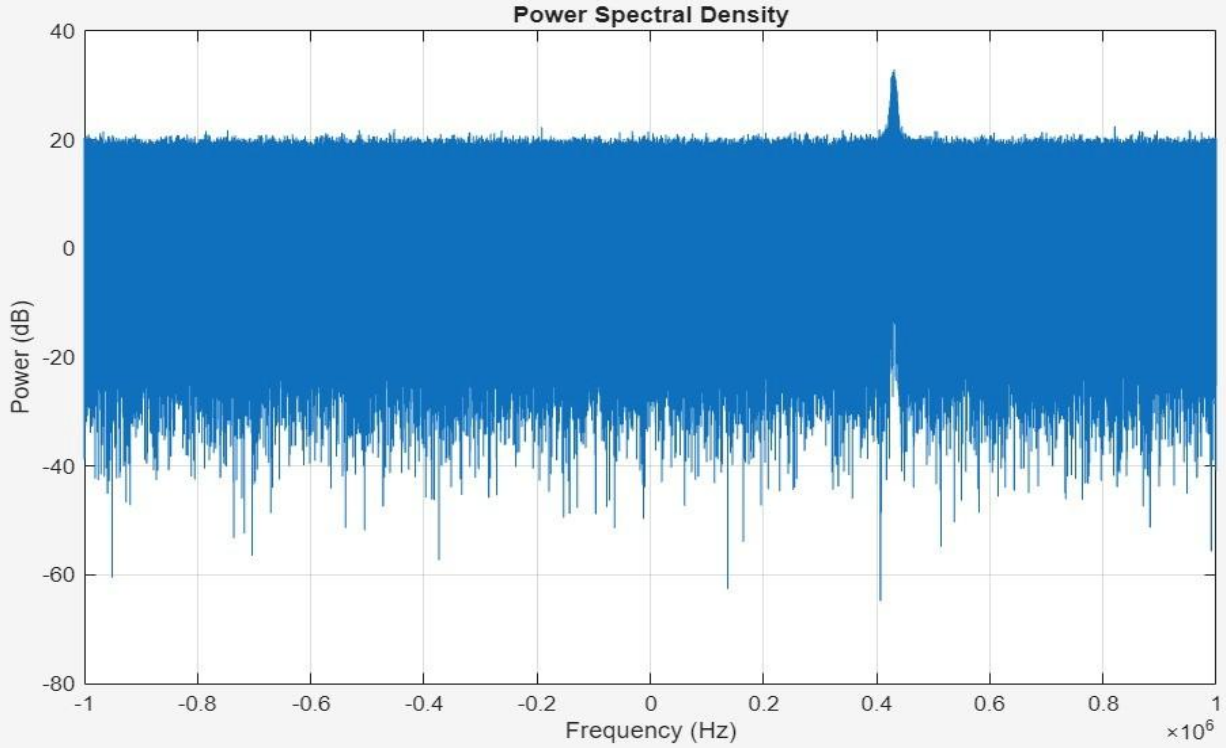


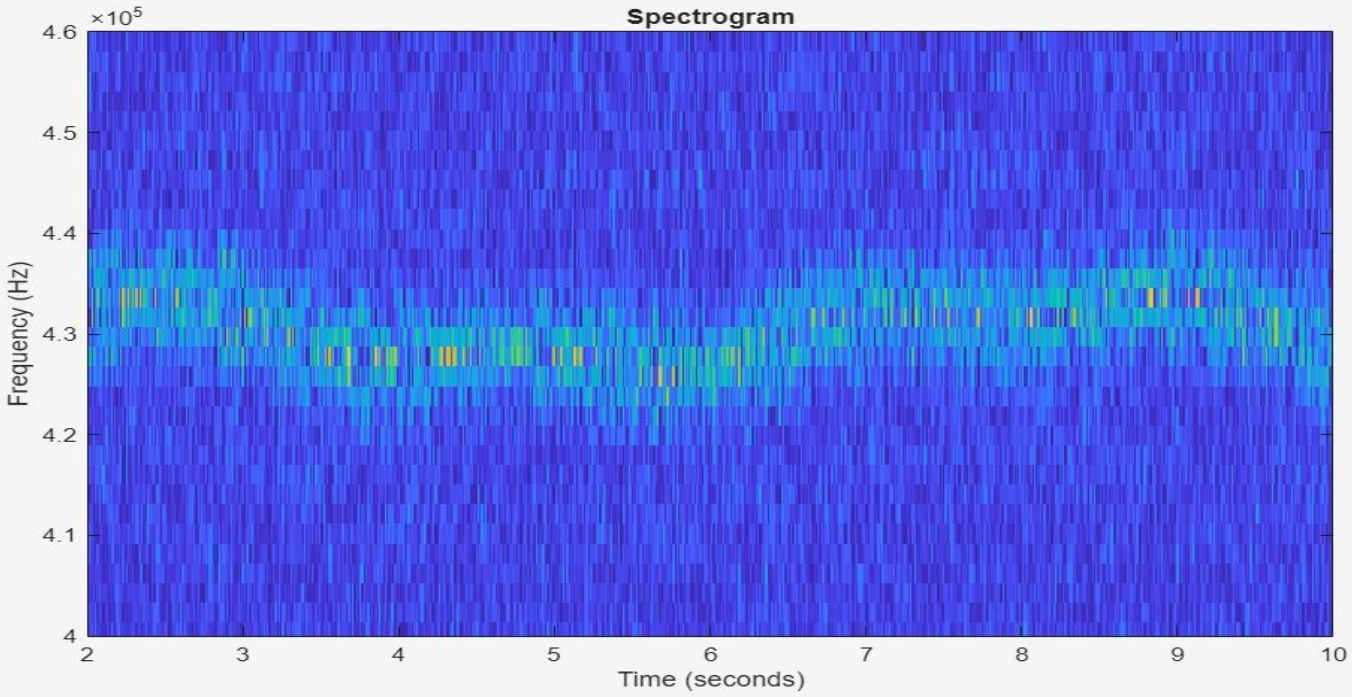
The IQ constellation diagram before correction shows the received symbols scattered around the origin instead of forming clear clusters. This spread indicates the presence of noise and channel distortions affecting the signal. As a result, symbol detection may be inaccurate and further correction is required to properly recover the transmitted signal. IQ constellation shows phase and amplitude distortions.

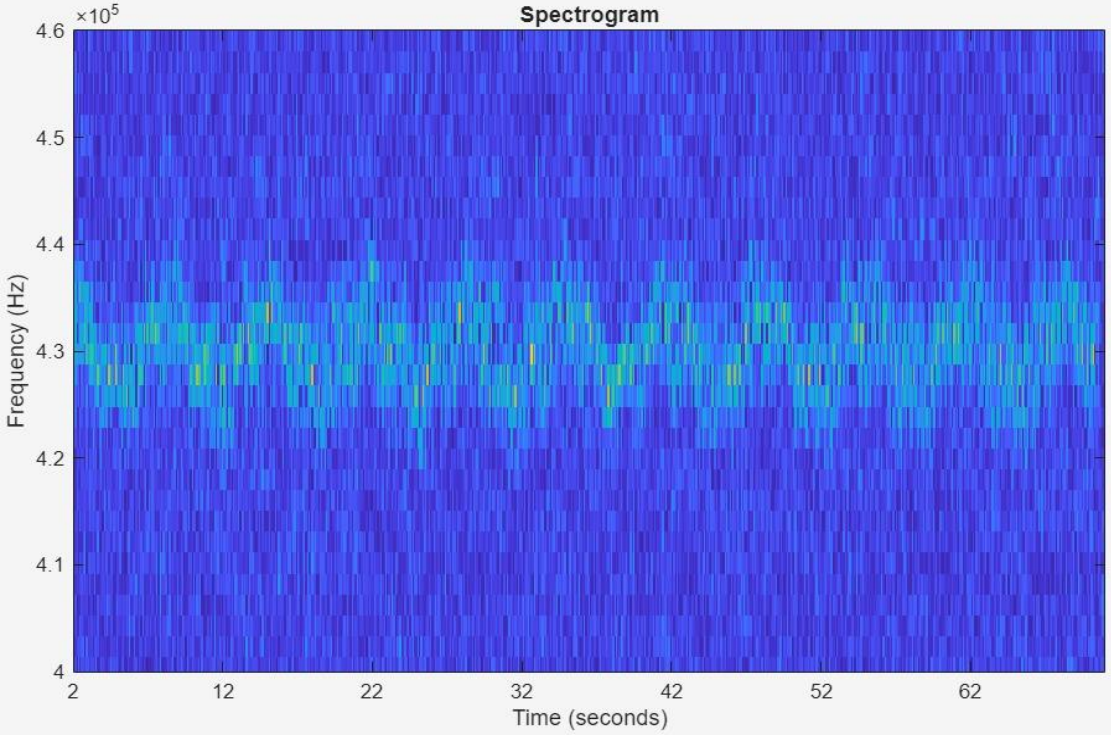


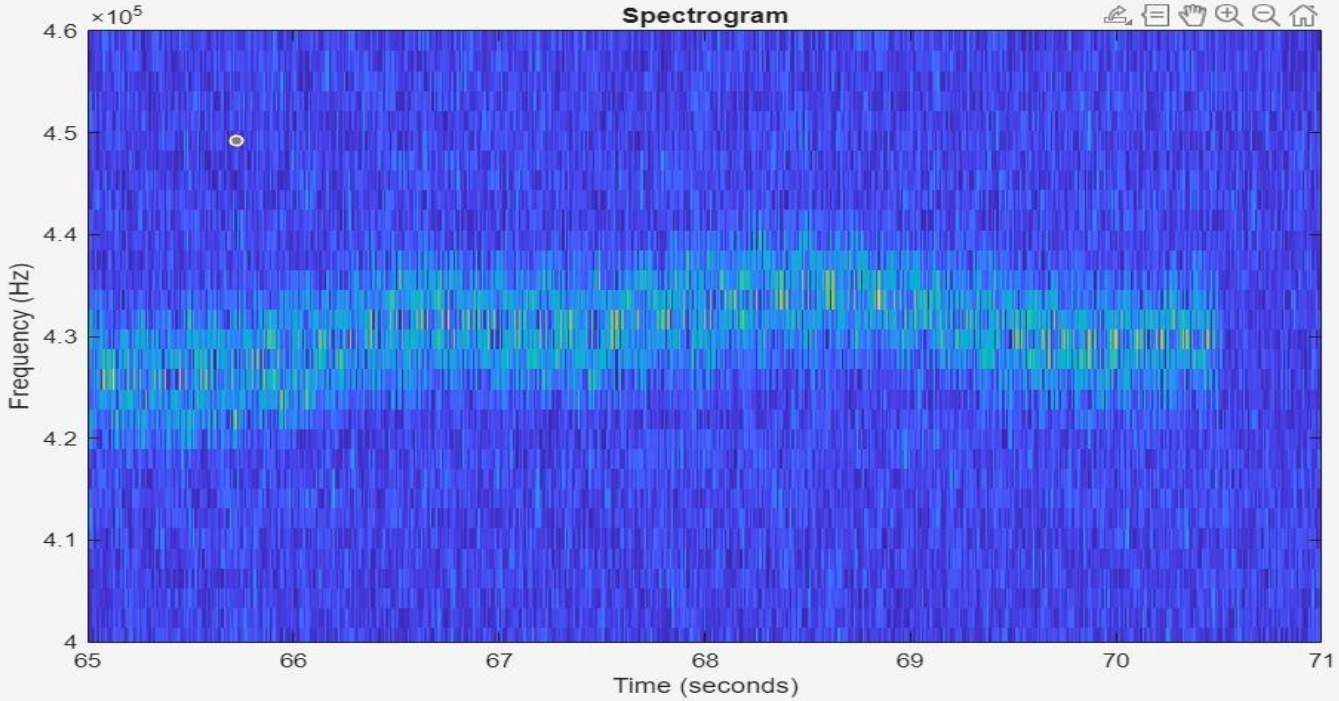

The Power Spectral Density (PSD) plot shows how the signal power is distributed across different frequencies. The spectrum appears mostly spread across the frequency range with a noticeable peak around a specific frequency, indicating a dominant signal component. The variations and lower-level fluctuations represent background noise present in the signal. This graph was obtained by using FFT(Fast Fourier Transform) to isolate the peak frequency.

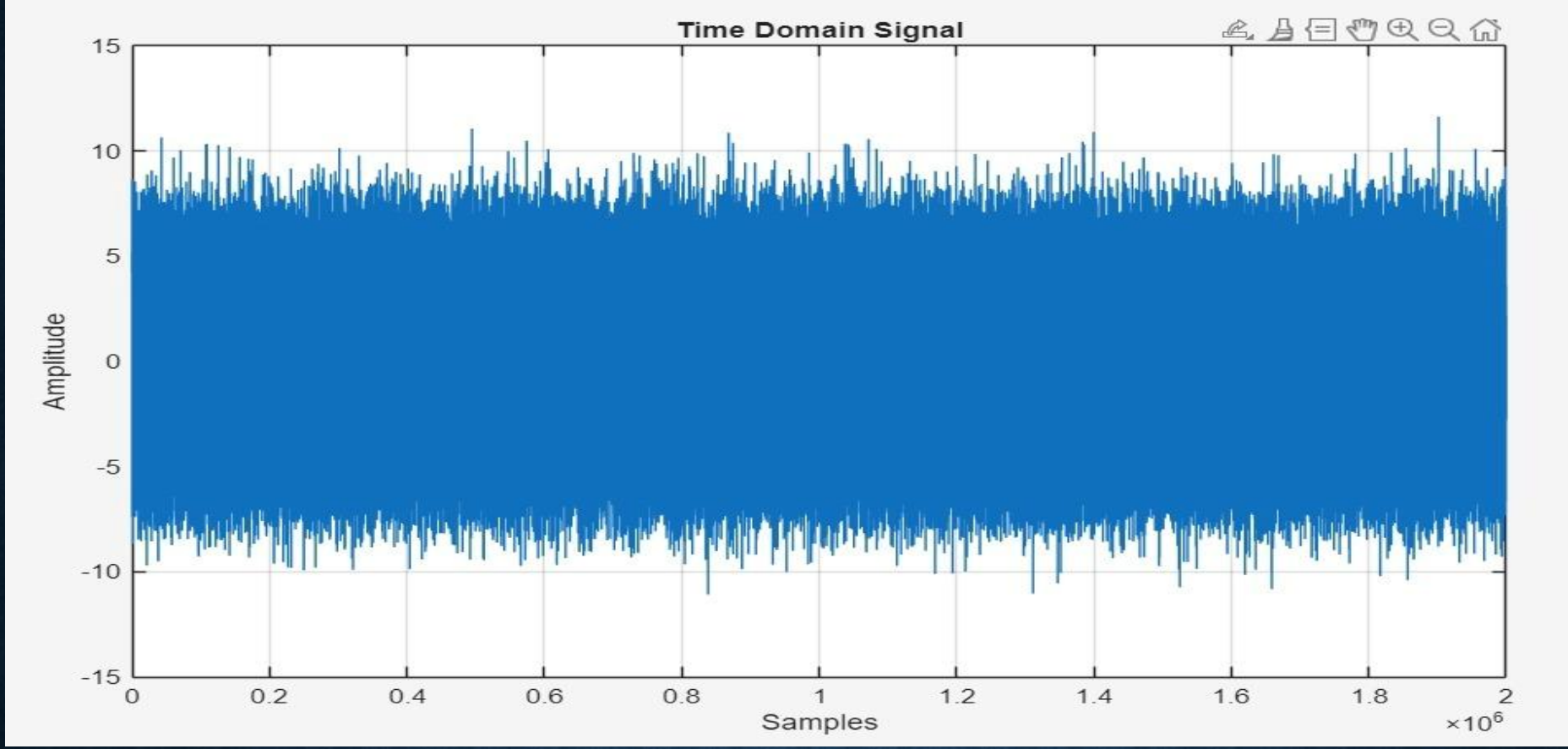


The spectrograms for files 3–10 and 66–71 show that the received carrier frequency drifts slightly up and down due to Doppler effects and oscillator instability, with the apparent frequency seen by the receiver deviating from the true carrier frequency transmitted by the probe. The third spectrogram is for all files from 3-71.






The time domain signal shows the variation of signal amplitude over the sampled time period. The waveform appears highly fluctuating, indicating the presence of noise along with the transmitted signal. The amplitude remains within a certain range, suggesting a stable signal with random variations due to channel or system noise.

# SIGNAL CHARACTERISTICS FROM INITIAL ANALYSIS

-**Time-domain signal:** Shows the raw received waveform containing modulation and noise.
-**Spectrogram:** Carrier frequency drifts slightly over time → indicates **Doppler shift and oscillator instability**.
-**Power Spectral Density:** Reveals the **apparent carrier frequency** observed by the receiver.
-**IQ Constellation (before correction):** Points are **rotated and spread**, indicating frequency and phase errors.

# PROBLEM IDENTIFIED

- The received signal is affected by:
- **Carrier Frequency Offset**

  Caused by **Doppler shift and oscillator drift**

  Observed as frequency shift in PSD and spectrogram
- **Phase / Polarization Rotation**

  Unknown antenna orientation causes constellation rotation
- **Clock Imperfections**

  Symbol rate may drift due to probe oscillator instability
- These distortions prevent direct symbol decoding.

# PROPOSED DSP CORRECTION PIPELINE

-To recover the transmitted data, the following steps are applied:

**-Carrier Frequency Estimation**

- Identify the carrier peak from the **power spectral density**

**-Frequency Offset Correction**

- Shift the signal spectrum to baseband

**-Carrier Phase Recovery**

- Rotate the constellation to align with the I/Q axes

**-Symbol Timing Recovery**

- Determine correct sampling instants for symbols

**-Constellation Detection**

- Map received points to nearest constellation symbols

# EFFECT OF SIGNAL CORRECTION AND BIT RECOVERY

- **Before Correction**

  -IQ constellation points appear **rotated and scattered**

  -Caused by **carrier frequency offset and phase rotation**

  -Symbols cannot be decoded reliably

- **After Frequency & Phase Correction**

  -Constellation points **cluster around ideal locations**

  -Signal aligns with **I/Q axes**

  -Enables accurate symbol detection

# SYMBOL TO BIT MAPPING AND DATA RECOVERY

- After correcting the carrier frequency and phase errors, the IQ constellation points align with the **ideal constellation locations**.
- Each constellation point corresponds to a **symbol**, which represents a specific **bit pattern**.
- **Example (QPSK Mapping)**

| CONSTELLATION POINTS | BITS |
|---|---|
| (+I,+Q) | 00 |
| (-I,+Q) | 01 |
| (-I,-Q) | 11 |
| (+I,-Q) | 10 |

# RESULT

-Correct constellation alignment enables **accurate symbol detection**
-Symbols are mapped to **binary bit sequences**
-This process reconstructs the **original bitstream transmitted by the probe**

# THANK YOU